

**FIRST SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020**

Mathematics

ME 1C 01—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

ME 1C 01—MATHEMATICAL ECONOMICS

(Multiple Choice Questions for SDE Candidates)

1. When the elasticity of demand e_p is greater than 1 the demand for good is ?
 - (A) Inelastic.
 - (B) Unitary elastic.
 - (C) Relatively elastic.
 - (D) Elastic.
2. When the cross price elasticity e_c is less than 0, the goods are ?
 - (A) Substitutes.
 - (B) Complementary.
 - (C) Independent.
 - (D) None of these.
3. The point price elasticity of demand e_p is given by :
 - (A) $\frac{dP}{dQ} \cdot \frac{P}{Q}$.
 - (B) $\frac{dQ}{dP} \cdot \frac{Q}{P}$.
 - (C) $\frac{dQ}{dP} \cdot \frac{P}{Q}$.
 - (D) $\frac{1}{P} \cdot \frac{dQ}{dP}$.
4. In the case of a perfectly inelastic supply curve, the elasticity of supply η_s is :
 - (A) Infinity.
 - (B) One.
 - (C) Greater than 1.
 - (D) Zero.
5. The variables in the demand function which are related to price are :
 - (A) Own price of the product.
 - (B) Price of compliment.
 - (C) Price of substitutes.
 - (D) All the above.
6. When the purchases of goods increase with rising levels of income, such goods are called :
 - (A) Inferior goods
 - (B) Normal goods.
 - (C) Giffen goods.
 - (D) Laxurious goods.
7. The influence of a change in a product's price on real income is called :
 - (A) Substitution effect.
 - (B) Income effect.
 - (C) Both (A) and (B).
 - (D) None.

8. In a demand curve price is measured along the :
- (A) Vertical axis. (B) Horizontal axis.
- (C) Both (A) and (B). (D) None.
9. When the elasticity of supply $\eta_p^s = 0$, the supply curve will be :
- (A) Parallel to x axis. (B) Passing through origin.
- (C) Parallel to y axis. (D) None.
10. For a unitary elastic supply curve, η_p^s is :
- (A) Less than 1. (B) More than 1.
- (C) Equal to 1. (D) Zero.
11. The price elasticity of demand of the demand function $Q = 400 - 4P$ at $p = 10$ is :
- (A) 0.11. (B) 0.10.
- (C) -0.11. (D) -1.
12. Luxury goods are :
- (A) Price inelastic. (B) Price elastic.
- (C) Both (A) and (B). (D) None.
13. The elasticity of demand η_d in terms of AR and MR is :
- (A) $\frac{AR - MR}{AR}$. (B) $\frac{AR - MR}{MR}$.
- (C) $\frac{MR}{AR - MR}$. (D) $\frac{AR}{AR - MR}$.
14. A distinction between cost of production and expenses of production is made by :
- (A) Engel. (B) Marshall.
- (C) Keynes. (D) None of these.

15. Total variable cost plus total fixed cost gives :

- (A) Total cost.
- (B) Average cost.
- (C) Marginal cost.
- (D) None of these.

16. A firm decide to discontinue production and accept a loss equal to it fixed cost :

- (A) Loss > FC.
- (B) Loss < FC.
- (C) Loss = FC.
- (D) None.

17. Which of the following is correct ?

- (A) Indifference curves slopes downward to the right.
- (B) Indifference curves do not intersect.
- (C) Indifference curves are convex to the origin.
- (D) All the above.

18. An attribute possessed by a commodity to satisfy a human want, to yield satisfaction to consumer is termed as :

- (A) Utility.
- (B) Preference.
- (C) Want.
- (D) None of these.

19. The second order condition for maximising utility is :

- (A) $\frac{du}{dq_1} = 0$.
- (B) $\frac{d^2u}{dq_1^2} > 0$.
- (C) $\frac{d^2u}{d^2q_1} < 0$.
- (D) $\frac{d^2u}{da_1^2} = 0$.

20. An indifference map is a collection of;

- (A) Indifference curves.
- (B) Cost curve.
- (C) Revenue curve.
- (D) None of these.

FIRST SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2020

Mathematics

ME 1C 01—MATHEMATICAL ECONOMICS

Time : Three Hours

Maximum : 80 Marks

Part A

*Answer all the **twelve** questions.*

Each question carries 1 mark.

1. The demand curve shows the relationship between :
 - (a) Price and quantity.
 - (b) Income and quantity.
 - (c) Consumption and quantity.
 - (d) Consumption and income.
2. The elasticity of demand at different points on the same demand curve is :
 - (a) Same.
 - (b) Zero.
 - (c) Different.
 - (d) None.
3. The total of the quantities demanded by all consumers in an economy at each price is called :
 - (a) Market demand curve.
 - (b) Market supply curve.
 - (c) Market equilibrium.
 - (d) None of the these.
4. Sum of explicit cost and implicit cost gives :
 - (a) Total cost.
 - (b) Average cost.
 - (c) Marginal cost.
 - (d) None of these.
5. The ratio of total cost to the quantity produced is called :
 - (a) Average cost.
 - (b) Marginal cost.
 - (c) Total variable cost.
 - (d) None.
6. When marginal cost is greater than average cost, the total cost elasticity will be :
 - (a) Greater than 1.
 - (b) Less than 1.
 - (c) Equal to 1.
 - (d) None.

Turn over

7. The concept of indifference curves was developed by :
- (a) J.R. Hicks. (b) R.G.D. Allen.
(c) J.R. Hicks and Allen. (d) None of these.
8. The equilibrium of a consumer purchasing one commodity is attained when :
- (a) $\frac{du}{dQ} < P$. (b) $\frac{du}{dQ} = P$.
(c) $\frac{du}{dQ} > P$. (d) $\frac{du}{dQ} = 0$.
9. The point at which the marginal utility first increases, reaches the maximum, then diminishes is called :
- (a) Point of inflexion. (b) Minimum point.
(c) Saturation point. (d) None of these.
10. Let $10 + 30kk^2$ be a production function, where k represents capital. Then the marginal productivity when $k = 3$ is :
- (a) 116. (b) 16.
(c) 58. (d) 24.
11. When the average revenue function is $AR = 10 - .5q$, the marginal revenue is :
- (a) $0.5q^2$. (b) $10 - q$.
(c) $10q0.5$. (d) 10.
12. Behaviour of the function defined by $y = x^4 - 6x^3 + 4x^2 - 13$ at $x = 4$ is :
- (a) Decreasing. (b) None.
(c) Increasing. (d) Stationary.

(12 × 1 = 12 marks)

Part B

*Answer any **six** questions in two **or** three sentences.*

Each question carries 3 marks.

13. What is a Demand Function ?
14. Write any three factors determining supply.
15. What is price elasticity of demand ?
16. Define average revenue and marginal revenue.
17. Define elasticity of total cost.
18. Write a note on Legrange's multiplier.
19. What do you mean by utility ?
20. Find average cost and marginal cost from the total cost function $TR = 10 + x + 2x^2$.
21. Show that the function $3x^3 + 3x^2 + x - 1$ is monotonic increasing.

(6 × 3 = 18 marks)

Part C

*Answer any **six** questions from the following.*

Each question carries 5 marks.

22. Describe various elasticities of demand.
23. What are the determinants of elasticity of demand ?
24. Give the nature and property of a demand function for a normal good.
25. Cost function is given by $\pi = a + bq + cq^2$. Prove that $\frac{d(AC)}{dq} = \frac{MC - AC}{q}$.
26. What are the similarities between utility approach and indifference curve approach ?
27. Find the maximum profit : Given the profit function $\pi = -Q^3 - 6Q^2 + 1440Q - 545$.
28. Find the critical points of $z = 3x^2 - xy + 2y^2 - 4x - 7y + 12$.

29. $z = \frac{x+y}{x+2}$. Find dz .

30. Find all the four second order partial derivatives of $z = 3x^3y^2$.

(6 × 5 = 30 marks)

Part D

Answer any **two** questions from the following.

Each question carries 10 marks.

31. Write short notes on determinants of price elasticity of demand.
32. (a) Prove that marginal cost (MC) must equal marginal revenue (MR) at the profit-maximizing level of output.
- (b) The total cost function of a firm is given by $TC = 400 - 10q + q^2$. Find the optimum size of output.
33. Explain briefly properties of indifference curves.
34. Given the profit function $\pi = 160x - 3x^2 - 2xy - 2y^2 + 120y - 18$ for a firm producing two goods x and y . Find the maximum profit.

(2 × 10 = 20 marks)